

Directors' Dealings Behavior of Managers of Top US High-Tech Firms Trading on Nasdaq After the US Presidential Election 2024: Event Study and ChatGPT/Mistral Validation

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ABSTRACT

Understanding how directors and high level managers trade their company's shares offers unique insight into insider confidence and perceived firm value, especially in periods of political transition. This report presents an event study of directors' dealings (DD) in 30 leading US high-tech firms following the US presidential election of 2024, focusing on purchase and sale events by executives. The study spans November 2024 to May 2025, examining market reactions via abnormal returns and cumulative abnormal returns around DD announcements using the Cross-Industry Standard Process for Data Mining (CRISP-DM) [9]. Statistically significant positive pre-event drift for both Sale ($CAR_{[-20,-1]} \approx +1.5\%$, $p = 0.038$) and Sale+OE events ($CAR_{[-20,-1]} \approx +2.5\%$, $p = 0.005$) was found, consistent with insiders timing transactions after periods of above normal stock performance, while post-event windows remain largely insignificant across all trade types. In addition, the report evaluates using a Large Language Model, specifically `mistral-small-3.2-24b`, to extract and process relevant event data as a starting point for prompt based financial analyses. The LLM-assisted event extraction yields hallucination rates exceeding 94%, and LLM-computed abnormal returns exhibit a systematic upward bias of 0.7–4.6% relative to the traditional OLS pipeline, indicating that small LLMs are not yet a reliable substitute for conventional event study methods.

1 INTRODUCTION

Directors' dealings (DD) are the purchase or sale of company shares by executives, managers, and other corporate insiders. They represent a unique window into management's private information and confidence in their firm's future. These transactions are mandated to be publicly disclosed under securities regulations and therefore provide researchers with observable signals of insider sentiment that may precede broader market recognition of fundamental value changes [1, 4, 5].

Studying directors' dealings becomes especially interesting during times of market uncertainty. The 2024 US presidential election is a good example of such a period as it brought significant uncertainty around tech regulation, trade policy, taxes, and innovation incentives. High-tech firms are particularly sensitive to these kinds of shifts, which makes them a good choice for examining how insider trading behavior changes in response to macroeconomic and political events.

Event study methods have been used for decades to detect abnormal returns around corporate announcements and insider trades [7]. The market model, going back to Sharpe [8] and later refined by Fama et al. [3], gives a solid way to separate firm-specific price

movements from general market trends. The main challenge has always been the amount of manual work involved, as collecting, cleaning, and analyzing data across many firms and time periods is time-consuming.

This is where Large Language Models (LLMs) like ChatGPT become interesting. These tools have shown impressive capabilities in acquiring information and performing analytical tasks, which raises the question of whether they could meaningfully and reliably speed up financial research workflows.

This project was carried out as part of the Interdisciplinary Project in Data Science course, which combines financial economics with data science methods. It looks at two connected questions. First, do directors' dealings in major US high-tech firms after the 2024 election generate statistically significant abnormal returns? This is a test of market efficiency and whether insider trades carry real informational content during politically uncertain times. Second, can an LLM reliably replicate the results of a traditional event study? For the initial exploration, ChatGPT-4 was used, but all final reported analyses use `mistral-small-3.2-24b` (accessed via the Aqueduct API provided by `datalab@tuwien`) to ensure reproducibility, as OpenAI's ChatGPT-4 non-deterministic Web-App Chat make exact replication impossible.

The study employs a comprehensive event study framework, analyzing directors' dealings across 30 major Nasdaq-listed technology companies from November 2024 through May 2025. Abnormal returns are calculated using the market model and examine cumulative abnormal returns across multiple event windows to capture pre-event drift as well as immediate and longer-term market reactions. Parallel analyses using LLM-extracted data allow for direct comparison of traditional versus AI-assisted research methodologies.

The remainder of this paper proceeds as follows: Section 2 outlines the experimental design and project setup, Section 3 details the CRISP-DM implementation framework including results and evaluation, Section 4 presents conclusions. The appendix provides supporting materials including the complete list of analyzed companies, a reference to the code generated and a description of all the data and prompts that were used.

2 EXPERIMENT DESIGN

The scope and setup of this project was provided by and aligned with Prof. Aussenegg and its original version can be found with all of the code and data used on Github [10]. This chapter outlines how the study is set up to investigate whether insider trading by

executives and directors in major tech companies creates measurable stock price reactions, and whether a LLM can reliably assist in collecting and analyzing this financial data.

Companies Sample: Thirty major technology companies listed on the Nasdaq stock exchange were selected. The complete company list is provided in Table 8 in the Appendix.

Study Period: The analysis runs from November 1, 2024 through May 31, 2025. This 7-month window allows both immediate reactions and longer-term market adjustments to insider trading to be captured during a period of political transition after the US presidential election.

Project Execution: The project was executed following the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework [9], which structures the workflow into six phases: business understanding, data understanding, data preparation, modeling, evaluation, and deployment. This methodology was chosen because it provides a clear structure for data driven research projects. The full CRISP-DM walkthrough is presented in Section 3, where each phase is documented in detail alongside the corresponding results.

Proposal Adjustments: ChatGPT-4 with internet access was used for initial exploratory experiments to assess feasibility, though the task of a full event study methodology was too complex and even with internet access, data extraction was insufficient. Therefore ChatGPT-4 was not used for the final quantitative analysis because its non-deterministic outputs and the absence of a fixed model version make exact replication impossible. All results are therefore based on `mistral-small-3.2-24b` hosted by the university. This approach nonetheless allows qualitative comparison of whether Mistral can match the quality of traditional data collection methods.

3 EXPERIMENT EXECUTION WITH CRISP-DM

This section closely follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) pipeline [9] to ensure methodological rigor and reproducibility.

3.1 Business Understanding

Before any data is collected or models are built, it is essential to clearly define what the study aims to achieve and why it matters. This section establishes the core research objectives, key terminology and success criteria that guide all subsequent phases.

3.1.1 Terminology. This subsection introduces the business terminology necessary for the event study methodology. Each term is introduced via a short definition:

Event Types: A “directors’ dealing event” is defined as any transaction where a corporate insider, such as a manager, executive officer, or major shareholder (owning more than 10% of company stock) – buys or sells company shares and derivative-only transactions and gift transfers are excluded. These are classified into three categories:

- **Purchase Events:** Insiders buying shares (potentially signaling confidence)
- **Sale Events:** Insiders selling shares (potentially signaling concerns or simply portfolio diversification)

- **Sale with Option Exercise (Sale+OE):** Sales preceded by the exercise of stock options, common in high-tech firms with equity-based compensation

Estimation Window: A period of 200 trading days $[-220, -21]$ prior to each event, used to estimate the market model parameters that define “normal” return behavior for a given stock.

Event Window: A set of trading days centered around the event date over which abnormal returns are measured. Multiple windows are used to capture different market reactions:

- (1) **Pre-event** $CAR_{[-20, -1]}$: Checks for information leakage before the official announcement
- (2) **Immediate** $CAR_{[0, 1]}$: Captures the announcement day and next-day reaction
- (3) **Short-term** $CAR_{[0, 5]}$: One-week response
- (4) **Medium-term** $CAR_{[0, 10]}$: Two-week adjustment periods
- (5) **Long-term** $CAR_{[0, 20]}$: Extended adjustment periods
- (6) **Full window** $CAR_{[-20, 20]}$: Captures the total effect across the entire 41-day window surrounding the event

Market Model: A regression based model that estimates the expected return of a stock as a linear function of the overall market return, used as the baseline for computing abnormal returns.

Abnormal Returns (AR): The difference between the observed return of a stock and its expected return as predicted by the market model on a given day.

Cumulative Abnormal Returns (CAR): The sum of all abnormal returns over a specific event window, capturing the total price impact associated with a directors’ dealing event.

3.1.2 Primary Objective. The primary objective is to determine whether directors’ dealings in 30 major US high-tech firms generate statistically significant abnormal returns in the post-election period (November 2024 – May 2025), and to evaluate whether Large Language Models can reliably replicate the traditional event study workflow. Both questions are addressed using the market model framework and non-parametric significance testing. Three main hypotheses are examined in this study:

H1 (Market Efficiency): If markets are efficient, directors’ dealings should *not* create significant abnormal returns because all public information is already reflected in stock prices. Testing this helps understand how well markets process insider trading information [4].

H2 (Information Content): It is expected that when insiders *buy* shares, stock prices should rise (positive abnormal returns), and when they *sell* shares, prices should fall (negative abnormal returns). This would demonstrate that insider trades carry valuable information about company prospects [1].

H3 (AI Methodology Validation): It is predicted that data extracted and analyzed using an LLM (`mistral-small-3.2-24b`) will produce statistically similar results to traditional manual approaches, which would validate the use of AI tools in financial research workflows.

3.1.3 Success Criteria. The project is considered successful if the following criteria are met:

- (1) **Statistical testing:** G-Rank significance tests are computed for all trade types and event windows, with clear rejection

or non-rejection of the null hypotheses at the 5% and 1% significance levels.

- (2) **LLM accuracy:** Paired comparison tests (t-test, Wilcoxon) between traditional and LLM-computed CARs show no statistically significant differences, validating H3.
- (3) **Reproducibility:** All analyses are fully scripted in Python and can be re-executed end-to-end from raw data.

3.2 Data Understanding

With the business objectives in place, the next step is to identify and explore the data needed to answer them. This section describes the primary data sources, how they were collected, and what initial inspection revealed about their structure and quality.

3.2.1 Data Sources. The study relies on two primary data sources to conduct the event study analysis and methodological comparison. A complete inventory of all input and generated data files is given in Table 9 in the Appendix.

OpenInsider.com provides comprehensive insider trading data sourced from SEC Form 4 filings (Table 1), which corporate insiders are legally required to submit within two business days of any transaction. The platform aggregates and structures this regulatory data, making it accessible for research purposes.

Table 1: Attribute details for `sec_filings.csv` (raw SEC Form 4 filings).

Column	Dtype	Non-Null
X	str	356/717
Filing Date	str	717/717
Trade Date	str	717/717
Ticker	str	717/717
Company Name	str	717/717
Insider Name	str	717/717
Title	str	717/717
Trade Type	str	717/717
Price	str	717/717
Qty	str	717/717
Owned	str	717/717
Δ Own	str	717/717
Value	str	717/717
1d	float64	0/717
1w	float64	0/717
1m	float64	0/717
6m	float64	0/717

Yahoo Finance serves as the source for individual stock returns (Table 2), the market benchmark (Nasdaq Composite Index, Table 3), and earnings call dates with surprise data (Table 10). Data is accessed programmatically through the `yfinance` Python library, which provides a reliable interface to Yahoo Finance's historical price database.

Table 2: Attribute details for `portfolio_returns.csv` (daily returns for all 30 portfolio stocks). One column per ticker; VMW has no data (delisted).

Column	Dtype	Non-Null
date	str	374/374
AAPL	float64	373/374
ADBE	float64	373/374
...	(26 more tickers)	373/374
VMW	float64	0/374
VZ	float64	373/374

Table 3: Attribute details for `market_returns.csv` (NASDAQ Composite daily returns).

Column	Dtype	Non-Null
date	str	373/374
market_return	float64	373/374

3.2.2 Data Collection. Director Dealing Collection Period: November 1, 2024 – May 31, 2025

This seven-month window encompasses the post-2024 US presidential election period, starting four calendar days before the election to establish pre-event baselines and extending through the end of May 2025 to capture medium-term market reactions.

Portfolio and Market Data Collection Period: December 1, 2023 - May 31, 2025 as 200 trading days before the first event window are needed.

Collection Method:

The data collection process follows a dual-track approach to enable methodological comparison:

Traditional Manual Collection:

- (1) **Directors' Dealings:** Insider transaction data is manually extracted from OpenInsider.com by filtering for the 30 selected companies (see Appendix Table 8) within the specified date range, copied and exported to csv via Excel.
- (2) **Company Stock and Market Price Data:** Daily price data and earnings call information for all 30 companies and the Nasdaq Composite Index is retrieved using the Python `yfinance` library.

AI-Assisted Collection:

- (1) `mistral-small-3.2-24b` (via Aqueduct API, no internet access) is prompted to extract insider trading data using the structured prompts described in Section 3.2.2, at both company-level and manager-level granularity.
- (2) Stock price data and market model parameters are computed by the LLM based on the same date ranges, using its parametric knowledge rather than live data retrieval.

LLM Prompting Strategy To evaluate LLM-based event extraction, two prompting strategies are tested using `mistral-small-3.2-24b` via the Aqueduct API (no internet access):

Company-level prompting: One prompt is issued per ticker, asking the model to recall all insider transactions for that company within the study period. This results in 30 prompts in total and represents the simpler, coarser approach. **Manager-level prompt-**

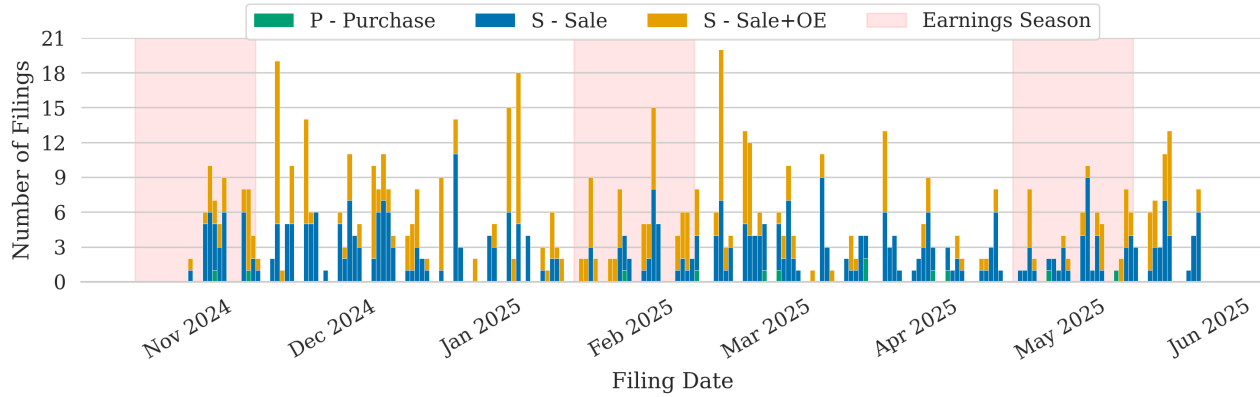


Figure 1: Daily distribution of Form 4 filings by trade type (November 2024 – May 2025). Shaded regions indicate approximate quarterly earnings seasons.

ing: One prompt is issued per (insider, ticker) pair, providing the model with the insider’s name as additional context. This yields 218 prompts and is designed to improve recall by narrowing the model’s focus to a specific individual.

In both cases, the model is asked to return structured output including the insider name, transaction date, transaction type (buy/sell), number of shares, and transaction value. The full text of all prompts is provided in Section 5 (LLM Prompts subsection).

Data Quality Issues Identified:

Several data quality challenges are encountered during the collection process:

- (1) **Stock Splits and Dividends:** Raw price data requires adjustment for corporate actions. The *yfinance* library provides adjusted close prices that account for splits and dividends, which are used for all return calculations to ensure consistency.
- (2) **Filing Date vs. Transaction Date Discrepancy:** SEC regulations allow up to two business days between transaction execution and public filing. This study uses the filing date (when the trade was publicly disclosed) as the event date, not the trade date.
- (3) **LLM Data Extraction Inconsistencies:** The AI-assisted collection method occasionally produces incomplete or incorrectly formatted data, particularly for complex transaction types. These discrepancies are documented as part of the methodological comparison (H3).

These quality issues are addressed through systematic data cleaning procedures documented in the Data Preparation phase (Section 3.3).

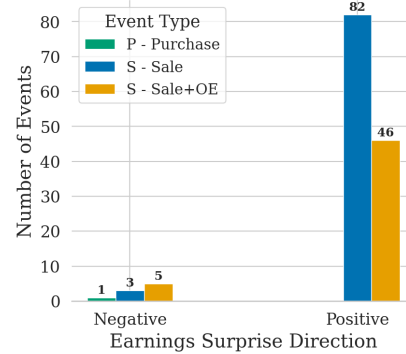


Figure 2: Barchart of insider trading activity relative to earnings announcement dates (≤ 20 days), highlighting potential clustering of directors’ dealings around earnings surprises.

3.2.3 Initial Data Exploration. Figure 1 shows the daily filing volume over the study period, stacked by trade type. Filing activity is spread unevenly across the seven months, with visible spikes during the quarterly earnings seasons (shaded). Several high-activity days coincide with post-earnings windows, consistent with insiders executing pre-planned trades shortly after blackout periods end.

The raw dataset contains 717 Form 4 filings across 29 of the 30 selected companies (VMW was delisted before the study period and produced no filings). After initial inspection, events are categorized into three transaction types: **Purchase** (“P – Purchase”), **Sale** (“S – Sale”), and **Sale with Option Exercise** (“S – Sale+OE”). Purchases are rare, accounting for only about 1.67% of events (12 samples), while sales and option-exercise sales dominate. Figure 4 shows the event count distribution across firms and transaction types. The concentration is stark, as just 6 firms account for 80% of all purchase events, 13 firms for 80% of sale events, and 10 firms for 80% of Sale+OE events, with META leading the sale category at over 50 events. This Pareto-like skew reflects the outsized insider trading activity of the largest-capitalization firms in the sample.

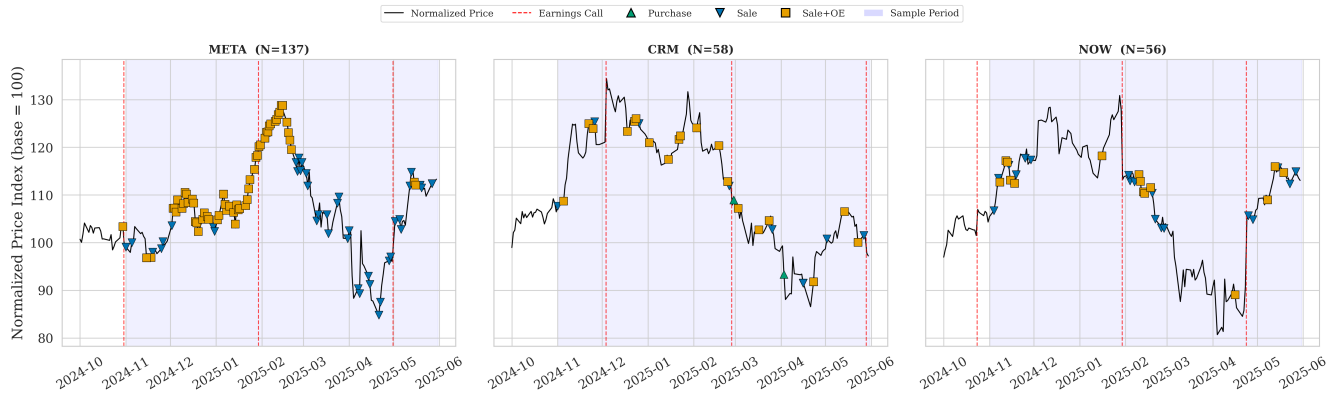


Figure 3: Stock price movements overlaid with insider trade events and quarterly earnings dates, illustrating the temporal clustering of directors' dealings around earnings periods.

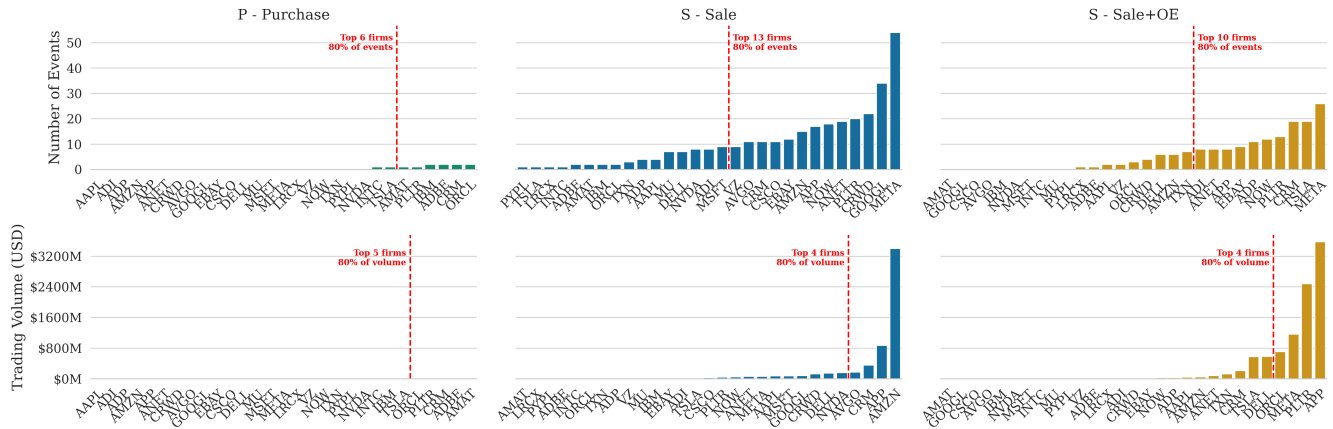


Figure 4: Top: Distribution of events by firm and transaction type (full sample of 717 events). Bottom: Trading volume (USD) by firm and transaction type, showing the concentration of dollar volume among the largest-capitalization firms.

Figure 4 reinforces this pattern in dollar terms, as just 4–5 firms account for 80% of total trading volume across all three transaction types, with AMZN dominating outright sales (exceeding \$3B) and PLTR and AAPL driving the bulk of option-exercise sale volume. Notably, the dollar concentration is even more pronounced than the event-count concentration, indicating that high-frequency filers are also high-value traders.

Figure 3 shows stock price movements overlaid with insider trades and earnings dates. Event dates are spread across the full November 2024 – May 2025 window, with notable clustering around quarterly earnings periods (see also Figure 2). The majority of events occur for the largest-capitalization firms (e.g., AAPL, MSFT, NVDA), while some smaller companies contribute only a handful of filings.

3.2.4 Data Quality Assessment. A systematic inspection of all four input datasets reveals several quality issues that must be addressed before modeling.

Insider filings (sec_filings.csv): The dataset contains no exact duplicate rows and no missing values in key fields (ticker, dates, insider name, trade type, price, quantity, value). However,

four return-performance columns (1d, 1w, 1m, 6m) are entirely null and are dropped. Numeric fields (price, quantity, value) are stored as formatted strings with dollar signs, commas, and sign prefixes (e.g., `-$945,233`) and require parsing. The filing lag between trade date and filing date averages 4.5 calendar days, with 80 events filed on the same day as the trade and 23 events filed more than 4 days late. One of the 30 target tickers (VMW) was delisted before the study period and produced no filings, leaving 29 active companies.

Market returns (market_returns.csv): Contains 374 trading days covering December 2023 through May 2025. The first row contains a null value (the return on the first date cannot be computed from a single closing price) and is dropped, leaving 373 valid daily returns.

Portfolio returns (portfolio_returns.csv): Contains daily returns for all 30 tickers over the same 374 day period. The VMW column is entirely null (delisted) and is excluded from analysis. All other tickers have 373 valid returns (the first row is null for the same reason as the market returns).

Earnings dates (earnings_dates_cache.csv): Contains 689 historical earnings records across 29 tickers. Two records are missing EPS estimates and one is missing the surprise percentage and these are retained since they only serve as supporting data for exploratory data analysis but don't affect results.

3.3 Data Preparation

With the quality issues identified above, the raw data is now filtered and cleaned to produce the final analytical sample.

3.3.1 Data Selection. From the 717 raw Form 4 filings, events are filtered to the 30 target tickers and the study period (November 2024 – May 2025). Only transaction types “P – Purchase”, “S – Sale”, and “S – Sale+OE” are retained. The all-null performance columns (1d, 1w, 1m, 6m) are dropped. Numeric fields (Price, Qty, Value) are parsed from formatted strings to floats. For the market and portfolio return data, the initial null row is removed and the VMW column is excluded.

3.3.2 Data Cleaning. Duplicate aggregation: When multiple insiders from the same company execute transactions of the same type on the same filing date, these are aggregated into a single event. This reduces the dataset from 717 raw filings to 492 unique (filing date, ticker, trade type) combinations – a reduction of 31.4%.

Conflict events: 36 (ticker, date) pairs have both a Sale and a Sale+OE transaction on the same day. Since these represent distinct transaction types with different economic interpretations (outright sale vs. option-exercise sale), both events are retained as separate entries, each assigned its own CAR in the event study.

3.3.3 Final Sample. After applying all filters and deduplication, the final sample contains **492 events** across 29 active tickers: 12 Purchase events (2.4%), 315 Sale events (64.0%), and 165 Sale+OE events (33.5%). The heavily skewed distribution reflects the well-documented tendency for insiders to sell far more frequently than they buy, especially in high-tech firms where stock-based compensation is prevalent.

3.4 Modeling

With a clean dataset in hand, the study applies the standard market model event study framework to quantify stock price reactions around directors' dealing events by calculating Abnormal Returns and Cumulative Abnormal Returns.

3.4.1 The Market Model. The standard market model from financial economics [2, 6, 7] is used to estimate what stock returns *should* be under normal conditions:

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \epsilon_{i,t} \quad (1)$$

where:

- $R_{i,t}$ = actual return on company i 's stock on day t
- $R_{m,t}$ = return on the overall market (Nasdaq Composite) on day t
- α_i = company-specific constant (how much the stock tends to move independently)
- β_i = sensitivity to market movements (how much the stock moves with the market)
- $\epsilon_{i,t}$ = random error term (unpredictable daily fluctuations)

Parameters α_i and β_i are estimated for each company using Ordinary Least Squares (OLS) regression on data from the estimation window.

3.4.2 Computing Abnormal Returns. Once what returns *should* be is known, abnormal returns (AR) are calculated as the difference between actual and expected [2, 6]:

$$AR_{i,t} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{m,t}) \quad (2)$$

where $\hat{\alpha}_i$ and $\hat{\beta}_i$ are the estimated parameters. A positive AR means the stock performed better than expected, while negative means worse.

3.4.3 Cumulative Abnormal Returns. To see the total effect over multiple days, daily abnormal returns are summed [2, 6]:

$$CAR_{i,[\tau_1, \tau_2]} = \sum_{t=\tau_1}^{\tau_2} AR_{i,t} \quad (3)$$

For example, $CAR_{[0,5]}$ adds up abnormal returns from the announcement day through the fifth day after, giving the cumulative impact over that week.

3.4.4 Model Execution. The modeling phase implements the event study as follows:

- (1) Estimation of α_i and β_i for each company via OLS on pre-event return data.
- (2) Calculation of ARs using the market-adjusted expected return model.
- (3) Replication of these calculations using a seeded LLM (mistral-small-3.2-24b) with no internet access (see Section 5, LLM Prompts subsection, for the exact prompt text).
- (4) Aggregation of ARs into CARs for the defined event windows.

3.5 Evaluation

Having computed the abnormal returns, the results must be rigorously tested to determine whether they are statistically meaningful and whether the LLM-assisted approach can match the traditional pipeline. This section presents descriptive statistics, non-parametric significance tests, and paired comparison tests across all trade types and event windows. Statistical testing includes:

- (1) Summary statistics (mean, median, standard deviation, minimum, maximum) for ARs and CARs across all six event windows for all groups: purchases and sales, both traditionally computed and LLM-computed. Full descriptive statistics are reported in Tables 4–6.
- (2) Non-parametric significance testing of the null hypotheses:

$$H_0 : \text{CAR-mean} = 0$$

$$H_0 : \text{CAR-median} = 0$$

using the Generalized Rank Test [6] as described in Section 3.5.1. The z-test variant is applied for Purchase events ($N < 30$) and the t-test variant for Sale and Sale+OE groups ($N \geq 30$).

- (3) Paired comparison tests between traditionally computed and LLM-computed CARs (H3), using the paired t-test and Wilcoxon signed-rank test as described in Section 3.5.2. Tests

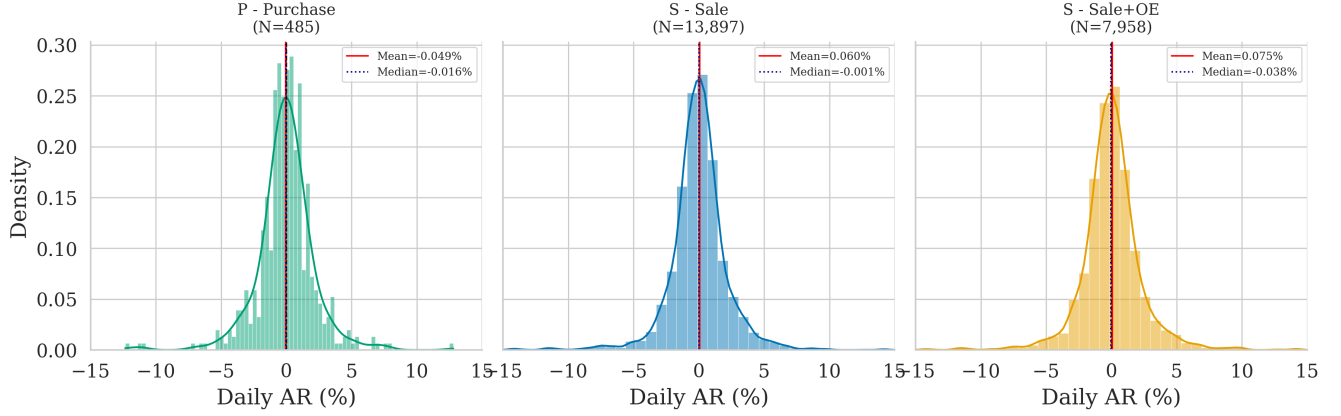


Figure 5: Daily abnormal return distributions by trade type, showing the non-normality at the individual-day level that motivates the use of non-parametric significance tests.

are conducted across all trade types (Purchase, Sale, Sale+OE) and all six event windows.

3.5.1 Non-Parametric Tests. To determine whether director dealings generate significant abnormal returns, the following null hypotheses are tested:

$$H_0 : CAR_{mean} = 0 \quad \text{and} \quad H_0 : CAR_{median} = 0 \quad (4)$$

Rejection of these hypotheses would provide evidence that directors' dealings contain value-relevant information that impacts stock prices. Given potential non-normality in financial returns [2], the Generalized Rank Test [6] is employed as the primary significance test. The procedure works as follows:

- (1) Compute standardized abnormal returns

$$SAR_{i,t} = AR_{i,t} / s_i \quad (5)$$

over the estimation window (length L_1) and

$$SCAR_i = CAR_i / (s_i \sqrt{L_2}) \quad (6)$$

for the event window (length L_2), where s_i is the residual standard deviation from the market model.

- (2) Cross-sectionally standardize:

$$SCAR_i^* = SCAR_i / S_{SCAR} \quad (7)$$

where S_{SCAR} is the cross-sectional standard deviation of $SCAR_i$.

- (3) Form the generalized series

$$GSAR_i = [SAR_{i,1}, \dots, SAR_{i,L_1}, SCAR_i^*] \quad (8)$$

and rank within each event to obtain

$$U_{i,t} = \text{rank}(GSAR_{i,t}) / (L_1 + 2) - 0.5 \quad (9)$$

- (4) Average the event-day rank across all N events: $\bar{U}_{L_1+1}$.

For $N \geq 30$, the test statistic follows a t -distribution with $L_1 - 1$ degrees of freedom:

$$t_{grank} = Z \sqrt{\frac{L_1 - 1}{L_1 - Z^2}}, \quad Z = \frac{\bar{U}_{L_1+1}}{S_{\bar{U}}} \quad (10)$$

where

$$S_{\bar{U}}^2 = (1/(L_1 + 1)) \sum_{t=1}^{L_1+1} \bar{U}_t^2 \quad (11)$$

For $N < 30$, a z -test is used with

$$S_{\bar{U}_{L_1+1}}^2 = L_1 / [12N(L_1 + 2)] \quad (12)$$

3.5.2 Cross-Sectional Analysis. To evaluate whether the LLM-assisted approach produces results comparable to the traditional OLS pipeline (H3), paired comparison tests are conducted between the two sets of CARs for each trade type and event window [2]. Two complementary tests are used:

Paired t-test: Tests whether the mean difference between traditionally computed and LLM-computed CARs is significantly different from zero. For each event i , the difference is defined as:

$$d_i = CAR_i^{\text{trad}} - CAR_i^{\text{LLM}} \quad (13)$$

The test statistic is:

$$t = \frac{\bar{d}}{s_d / \sqrt{N}} \quad (14)$$

where $\bar{d}$ is the mean difference across all N events and s_d is the standard deviation of the differences. Under the null hypothesis

$$H_0 : \bar{d} = 0 \quad (15)$$

, the statistic follows a t -distribution with $N - 1$ degrees of freedom. This parametric test assumes approximate normality of the differences but is reasonably robust for large samples.

Wilcoxon Signed-Rank Test: A non-parametric alternative that tests whether the median difference between the two sets of CARs is zero, without assuming normality. The differences d_i are ranked by absolute value, and the test statistic is:

$$W = \sum_{i=1}^N \text{sgn}(d_i) \cdot R_i \quad (16)$$

where

$$R_i = \text{rank}(|d_i|) \quad (17)$$

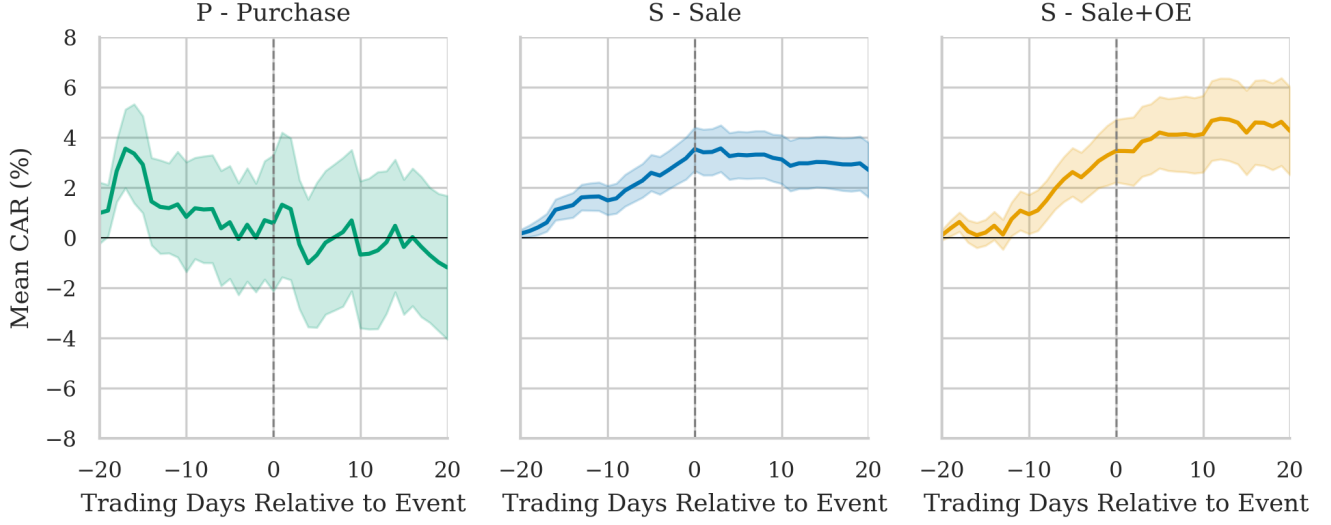


Figure 6: Cumulative abnormal return trajectories around event day 0, faceted by trade type. The shaded region indicates the event window and the vertical dashed line marks the event date.

is the rank of the absolute difference and $\text{sgn}(d_i)$ is its sign. Under the null hypothesis

$$H_0 : \text{median}(d_i) = 0 \quad (18)$$

, W is approximately normally distributed for $N \geq 25$:

$$z = \frac{W}{\sqrt{N(N+1)(2N+1)/6}} \quad (19)$$

This test is particularly useful given the potential for skewed CAR distributions and serves as a robustness check on the paired t-test results.

Both tests are applied across all trade types (Purchase, Sale, Sale+OE) and all six event windows. A statistically significant result in either test would indicate that the LLM-computed CARs deviate systematically from the traditional estimates, providing evidence against H3. Conversely, if both tests are insignificant across all windows, this would support the use of LLMs as a reliable substitute for traditional event study computations.

3.5.3 Descriptive Statistics. The mean and median CARs across event windows for each trade type are as follows:

- **Purchase (N=12):** Mean CARs are small and negative across all windows (e.g., mean $\text{CAR}_{[-20,20]} \approx -3.3\%$, median $\approx -7.5\%$). The tiny sample size precludes any meaningful interpretation.
- **Sale (N=315):** Mean CARs are close to zero across post-event windows, with $\text{CAR}_{[0,20]} \approx -0.7\%$, while a statistically significant positive pre-event drift of $\text{CAR}_{[-20,-1]} \approx +1.5\%$ ($p = 0.038$) is observed.
- **Sale+OE (N=165):** This group shows a distinctive pattern: a **positive pre-event drift** of approximately $+2.5\%$ in the $[-20, -1]$ window ($p = 0.005$), suggesting price appreciation in the weeks leading up to option-exercise sales. The full-window $\text{CAR}_{[-20,20]}$ reaches $+2.8\%$ ($p = 0.011$).

Tables 4–6 present full descriptive statistics for all groups across all CAR windows.

3.5.4 G-Rank Significance Tests. The Generalized Rank Test [6] is used to test $H_0 : \text{mean CAR} = 0$ and $H_0 : \text{median CAR} = 0$ for each trade type and event window. The z-test variant is used for Purchase events ($N < 30$) and the t-test variant for Sale and Sale+OE groups ($N \geq 30$).

With only 12 Purchase events, statistical power is extremely low. Most windows show no significant results (most p-values > 0.20). However, the full-window $\text{CAR}_{[-20,20]}$ reaches $p = 0.033$ (significant at 5%), driven by a cumulative negative drift of -3.3% . Given the very small sample size, this result should be interpreted with caution.

Figure 6 shows the CAR trajectories around event day 0 for each trade type. The Sale group ($N=315$) shows a **statistically significant positive pre-event drift**. $\text{CAR}_{[-20,-1]}$ reaches $p = 0.038$ (significant at 5%), with a mean of $+1.5\%$. Post-event windows remain insignificant: $\text{CAR}_{[0,20]}$ has $p = 0.156$ and $\text{CAR}_{[-20,20]}$ has $p = 0.442$. The mean post-event drift is negligible at approximately -0.7% over 20 trading days. This pre-event drift is consistent with insiders selling after a period of sustained outperformance.

Option-exercise sales show the study's strongest finding: a **robust positive pre-event drift**, with two windows reaching statistical significance. For Sale+OE events ($N=165$), $\text{CAR}_{[-20,-1]}$ has $p = 0.005$ (significant at 1%) and $\text{CAR}_{[-20,20]}$ has $p = 0.011$ (significant at 5%). The estimated pre-event drift is approximately $+2.5\%$ in the $[-20, -1]$ window, with the full window $\text{CAR}_{[-20,20]}$ reaching $+2.8\%$. This pattern is consistent with information leakage or anticipatory trading, as insiders may time their option exercises and sales to coincide with periods of positive stock performance, suggesting they exploit private knowledge about near-term price movements.

Table 4: Descriptive statistics, G-Rank tests, and paired comparison for Purchase events: Group 5a (Traditional, $N = 12$) vs. Group 5c (LLM, $N = 12$). All values in %. G-Rank uses Z-test ($N < 30$); $p_\mu/p_{\bar{x}}$: mean/median test. Δ = Trad-LLM; p_t : paired t-test; p_W : Wilcoxon. * $p < 0.05$; ** $p < 0.01$.

Window	5a: Traditional					5c: LLM					G-Rank		Trad – LLM			
	Mean	Med	Std	Min	Max	Mean	Med	Std	Min	Max	p_μ	$p_{\bar{x}}$	Δ Mean	Δ Med	p_t	p_W
$CAR_{[-20,-1]}$	-1.89	-3.91	6.54	-13.30	10.36	-0.76	-2.18	7.32	-14.02	14.50	0.231	0.588	-1.13	-1.73	0.611	0.569
$CAR_{[0,1]}$	0.30	-0.85	2.78	-2.63	6.47	0.78	-0.15	2.83	-1.87	7.85	0.329	0.219	-0.48	-0.71	0.423	0.677
$CAR_{[0,5]}$	-0.90	-0.25	3.73	-8.05	3.69	0.56	0.47	3.40	-4.54	7.30	0.608	0.952	-1.46	-0.72	0.079	0.129
$CAR_{[0,10]}$	-1.13	0.26	5.67	-11.91	6.38	0.45	-0.16	4.36	-6.80	7.00	0.976	0.716	-1.58	+0.42	0.134	0.129
$CAR_{[0,20]}$	-1.45	-2.49	5.04	-9.84	5.75	1.39	0.33	4.91	-5.96	9.38	0.320	0.577	-2.84	-2.82	0.137	0.204
$CAR_{[-20,20]}$	-3.34	-7.49	9.47	-14.33	14.84	0.63	-0.67	8.93	-8.61	19.27	0.033*	0.212	-3.97	-6.82	0.298	0.339

Table 5: Descriptive statistics, G-Rank tests, and paired comparison for Sale events: Group 5b (Traditional, $N = 315$) vs. Group 5d (LLM, $N = 315$). All values in %. G-Rank uses t-test ($N \geq 30$); see Table 4 for column definitions.

Window	5b: Traditional					5d: LLM					G-Rank		Trad – LLM			
	Mean	Med	Std	Min	Max	Mean	Med	Std	Min	Max	p_μ	$p_{\bar{x}}$	Δ Mean	Δ Med	p_t	p_W
$CAR_{[-20,-1]}$	1.46	1.07	11.27	-42.33	62.33	3.27	1.45	15.60	-45.88	144.04	0.038*	0.951	-1.81	-0.39	0.000**	0.001**
$CAR_{[0,1]}$	-0.01	-0.06	3.35	-14.17	24.87	-0.06	-0.07	4.03	-24.32	27.85	0.909	0.703	+0.05	+0.01	0.644	0.151
$CAR_{[0,5]}$	-0.24	-0.16	4.86	-19.45	26.97	-0.25	-0.04	5.79	-29.04	31.39	0.824	0.486	+0.01	-0.12	0.935	0.290
$CAR_{[0,10]}$	-0.44	-0.40	6.52	-34.40	42.74	-0.18	-0.10	7.89	-35.60	46.67	0.241	0.634	-0.26	-0.30	0.231	0.039*
$CAR_{[0,20]}$	-0.69	-0.21	7.51	-29.03	36.57	-0.01	-0.17	8.94	-36.60	43.12	0.156	0.403	-0.69	-0.04	0.012*	0.019*
$CAR_{[-20,20]}$	0.77	-0.06	12.65	-39.90	68.99	3.26	0.31	18.19	-38.61	142.50	0.442	0.366	-2.50	-0.37	0.000**	0.000**

Table 6: Descriptive statistics, G-Rank tests, and paired comparison for Sale+OE events: Group 5b' (Traditional, $N = 165$) vs. Group 5d' (LLM, $N = 165$). All values in %. G-Rank uses t-test ($N \geq 30$); see Table 4 for column definitions.

Window	5b': Traditional					5d': LLM					G-Rank		Trad – LLM			
	Mean	Med	Std	Min	Max	Mean	Med	Std	Min	Max	p_μ	$p_{\bar{x}}$	Δ Mean	Δ Med	p_t	p_W
$CAR_{[-20,-1]}$	2.49	2.13	11.92	-26.34	68.22	5.62	3.53	18.60	-29.11	144.04	0.005**	0.440	-3.13	-1.40	0.000**	0.000**
$CAR_{[0,1]}$	-0.30	-0.16	3.23	-12.73	14.14	-0.14	-0.15	4.35	-24.32	13.99	0.352	0.706	-0.16	-0.02	0.343	0.082
$CAR_{[0,5]}$	-0.42	-0.04	5.61	-21.23	20.06	-0.22	-0.08	6.93	-29.04	23.48	0.339	0.410	-0.20	+0.04	0.402	0.109
$CAR_{[0,10]}$	-0.00	-0.02	6.52	-17.31	26.54	0.62	-0.09	8.63	-29.17	38.43	0.724	0.741	-0.62	+0.07	0.053	0.026*
$CAR_{[0,20]}$	0.29	0.05	7.95	-22.89	32.07	1.79	0.03	9.90	-36.60	36.08	0.980	0.932	-1.51	+0.03	0.000**	0.001**
$CAR_{[-20,20]}$	2.77	0.53	14.43	-37.09	51.37	7.41	4.10	22.20	-37.51	142.50	0.011*	0.149	-4.64	-3.57	0.000**	0.000**

All median tests are **insignificant** across every trade type and window, indicating that the significant mean results are tail-driven rather than reflecting a symmetric shift in the CAR distribution (see Figures 7 and 5). A small number of large positive CARs pull the mean, while the typical event shows a CAR near zero.

3.5.5 LLM Validation: Traditional vs. AI-Computed CARs. To test H3, CARs computed from the traditional OLS pipeline are compared with those produced by `mistral-small-3.2-24b` using paired t-tests and Wilcoxon signed-rank tests. For the 12 Purchase events, **no significant paired differences** are found in any window. Both tests are insignificant (all $p > 0.07$), confirming that traditional and LLM-computed CARs agree well for this trade type.

The Sale group ($N=315$) reveals **significant divergences**, particularly in windows that include the pre-event period. $CAR_{[-20,-1]}$ shows paired t-test $p < 0.001$ and Wilcoxon $p < 0.001$, with a mean difference of -1.8% (LLM higher). $CAR_{[-20,20]}$ yields a mean difference of -2.5% (both tests $p < 0.001$), $CAR_{[0,20]}$ a mean difference of -0.7% (t-test $p = 0.012$, Wilcoxon $p = 0.019$), and $CAR_{[0,10]}$ reaches

significance in the Wilcoxon test ($p = 0.039$). The immediate-reaction windows ($CAR_{[0,1]}$ and $CAR_{[0,5]}$) remain insignificant in both tests, indicating that the methods agree on short-term event-day returns but diverge over longer horizons where differences in market-model parameter estimates ($\hat{\alpha}$, $\hat{\beta}$) accumulate.

The Sale+OE group ($N=165$) shows the most pronounced divergence, with both tests reaching significance across multiple windows: $CAR_{[-20,-1]}$ (mean difference -3.1%, both $p < 0.001$), $CAR_{[0,20]}$ (mean difference -1.5%, both $p < 0.001$), $CAR_{[-20,20]}$ (mean difference -4.6%, both $p < 0.001$), and $CAR_{[0,10]}$ (Wilcoxon $p = 0.026$). These divergences are not merely outlier-driven: the Wilcoxon tests – which are robust to outliers – also reject the null of equal distributions, indicating a systematic shift rather than a few extreme cases.

The LLM introduces a consistent **upward bias** of 0.7–4.6% relative to the traditional OLS approach. This bias is most pronounced for Sale+OE events and in windows spanning the estimation period

boundary, likely because the LLM’s market-model parameter estimates ($\hat{\alpha}_{LLM}$, $\hat{\beta}_{LLM}$) differ systematically from the OLS estimates and these differences compound over multi-day CAR windows. **H3 is therefore not supported.** *mistral-small-3.2-24b* should not be treated as a substitute for proper OLS estimation.

3.5.6 LLM Event Extraction: Accuracy Evaluation. A separate experiment evaluates whether *mistral-small-3.2-24b* can extract directors’ dealing events from parametric memory alone, using the two prompting strategies described in Section 3.2.2. Extracted events are matched against the SEC filings ground truth (717 events) using fuzzy name matching, a ± 3 calendar-day date tolerance, and trade-direction agreement. Table 7 reports the aggregate results for both approaches.

Table 7: LLM event extraction accuracy (*mistral-small-3.2-24b*), comparing company-level and manager-level prompting strategies.

Metric	Manager	Company
Total extracted	1,176	212
Ground truth events	717	717
True positives	67	5
False positives	1,109	207
False negatives	650	712
Precision	0.057	0.024
Recall	0.093	0.007
F1 Score	0.071	0.011
Hallucination rate	94.3%	97.6%
Date accuracy (matched)	30.2%	20.0%
Type accuracy (matched)	61.4%	60.0%

The results are clearly negative for both approaches. The manager-level strategy, which provides the model with specific insider names, performs better but remains far from usable: of the 1,176 events returned, only 67 (5.7%) correspond to real filings, and of the 717 ground-truth events the model recovers only 67 (9.3%), missing 650. Even among correctly matched events, only 30% have the exact trade date right and the mean value error exceeds 100%, indicating that the model produces plausible-sounding but largely incorrect transaction details.

The company-level approach fares even worse: with only 212 extracted events and 5 true positives, precision drops to 2.4% and recall to 0.7%, yielding a hallucination rate of 97.6%. The coarser prompting strategy, which asks the model to recall all insider transactions for a given company without specifying individual names, clearly provides insufficient context for the model to retrieve accurate structured data.

Performance varies sharply across tickers: 12 of 29 tickers (including NVDA, ADBE, IBM, and INTC) yield zero true positives in the manager-level approach, while META, PLTR, and NOW each produce 10–11 matches — likely because these companies had more prominent insider trading coverage in the model’s training data.

These findings make clear that current LLMs cannot reliably extract structured financial event data from parametric memory.

The hallucination rate of $>94\%$ makes unsupervised use of LLM-extracted events infeasible for event study research. Combined with the CAR computation divergences above, this indicates that LLMs face real limitations in both factual recall and precise numerical computation for this kind of workflow.

3.5.7 Summary of Findings. Returning to the three hypotheses:

H1 (Market Efficiency): Results are mixed. Purchase events show a significant negative $CAR_{[-20,20]}$ ($p = 0.033$), but the tiny sample ($N=12$) limits confidence. Sale events show a significant positive pre-event drift of $+1.5\%$ ($p = 0.038$), while post-event windows are insignificant. Sale+OE events show a robust positive pre-event drift — significant at 1% for $CAR_{[-20,-1]}$ ($p = 0.005$) and at 5% for $CAR_{[-20,20]}$ ($p = 0.011$) — suggesting that markets do not fully anticipate option-exercise sales.

H2 (Information Content): The expected direction — positive CARs for purchases, negative for sales — is only partially supported. Purchase CARs are negative (opposite to prediction, though the full-window result is significant). Sale and Sale+OE post-event CARs are near zero and insignificant. The strongest signals are the pre-event drifts for Sale and Sale+OE, consistent with insiders selling after periods of above-normal performance. This contrasts with Gębka et al. [5], who find that insiders in the majority of European countries do not generate statistically significant abnormal trading profits.

H3 (AI Validation): Traditional and LLM-computed CARs diverge significantly for both Sale and Sale+OE events. Paired t-tests and Wilcoxon tests both reject equality for pre-event and full-window CARs, indicating a systematic upward bias of 0.7–4.6% in LLM estimates. Only Purchase events ($N=12$) show no significant differences. The event extraction experiment further reveals that LLMs cannot reliably retrieve financial event data from memory (precision 5.7%, recall 9.3%, hallucination rate $>94\%$). H3 is **not supported**.

3.5.8 Result Limitations. While the study is designed for rigor and reproducibility, several limitations are worth acknowledging.

Directors’ dealings databases such as OpenInsider may miss or delay certain transaction filings, introduce errors, or use inconsistent definitions that are hard to detect, which could affect the completeness and accuracy of the event sample. On the modeling side, the market model assumes linear relationships between stock and market returns and constant parameters over the event window, meaning that short-term market anomalies, bid-ask spread effects, and overlooked confounding variables could bias the calculated abnormal returns. The selection of ± 20 days for the event window also involves a trade-off: too narrow a window may miss longer-term effects, while too broad a window increases noise and the risk of unrelated events contaminating the results. Additionally, merging multiple transactions by different individuals on the same day for a given company and event type into a single event may conflate distinct information signals and obscure more nuanced trading patterns. Finally, with only 29 firms and a relatively concentrated event sample, statistical inference is limited and the findings may not generalize beyond high-tech Nasdaq firms, as different market structures, DD regulations, and sector characteristics could lead to quite different results in other contexts.

The ability of `mistral-small-3.2-24b` to accurately retrieve relevant directors' dealings is constrained by the quality and recency of its training data, the precision of the prompts used, and the fundamental limitation of operating without web access — all of which severely reduce factual recall of specific transaction records. The model also does not perform real-time data validation, which can introduce inconsistencies relative to traditional sources. The high hallucination rate (94.3%) observed in the event extraction experiment underscores this risk, and suggests that a more recently trained or larger model might yield meaningfully better results.

These limitations should be kept in mind when interpreting the results. They also point toward useful directions for future research and highlight the importance of robust validation — particularly as LLMs take on a growing role in empirical finance.

3.6 Deployment

"Deployment" is treated as the compilation of full scientific documentation, publication of code, data, and results, and preparation of a scientific poster. The project is designed for reproducibility and all analyses are open-source and can be rerun with access to the data and scripts.

All code and data are publicly available on GitHub [10].

4 CONCLUSION

This project examined directors' dealings (DD) in 30 major Nasdaq-listed US high-tech firms over a seven-month post-2024 presidential election window (November 2024 – May 2025), applying a standard market model event study framework to 492 qualifying events across three transaction types: purchases, sales, and option-exercise sales. The study was carried out as part of the Interdisciplinary Project in Data Science course, and the entire analysis pipeline was structured and presented using the CRISP-DM framework, showing how a data mining methodology can be applied to a financial economics research question in practice. Alongside the traditional OLS pipeline, a parallel LLM-assisted workflow using `mistral-small-3.2-24b` was evaluated — both for computing CARs and for extracting event data from parametric memory — to explore whether AI tools can meaningfully support or replicate financial event study research.

The main empirical finding is a robust positive pre-event drift for Sale+OE events ($CAR_{[-20,-1]} \approx +2.5\%$, $p = 0.005$), consistent with insiders timing option-exercise sales to coincide with periods of above-normal stock performance. Sale events show a similar but weaker pre-event drift ($+1.5\%$, $p = 0.038$), while post-event windows remain largely insignificant across all trade types. The LLM validation produced clearly negative results, as systematic CAR divergences of 0.7–4.6% were found for Sale and Sale+OE groups, and the event extraction experiment revealed a hallucination rate exceeding 94% (precision 5.7%, recall 9.3%), making unsupervised LLM-based event data collection infeasible in its current form.

Several directions for future work follow from these findings. Extending the sample to more firms and a longer time horizon would improve statistical power, especially for the underpowered purchase category. Comparing multiple LLM architectures, more recent models and prompt strategies would help clarify whether the

observed upward bias is model specific or a more general LLM limitation. Most promisingly, retrieval-augmented generation (RAG) pipelines could pair LLMs with structured SEC filing databases rather than relying on parametric memory that could address the factual recall problem and bring AI-assisted event study workflows meaningfully closer to practical viability.

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5 APPENDIX

5.1 Company Sample

Table 8: Nasdaq Technology Company Tickers

#	Company Name	Ticker
1	Microsoft	MSFT
2	NVIDIA	NVDA
3	Apple	AAPL
4	Amazon	AMZN
5	Alphabet (Google)	GOOGL
6	Meta Platforms	META
7	Tesla	TSLA
8	Broadcom	AVGO
9	Oracle	ORCL
10	Palantir Technologies	PLTR
11	Salesforce	CRM
12	CrowdStrike	CRWD
13	Arista Networks	ANET
14	AppLovin	APP
15	Intel	INTC
16	IBM	IBM
17	Adobe	ADBE
18	PayPal	PYPL
19	ServiceNow	NOW
20	Cisco Systems	CSCO
21	Dell Technologies	DELL
22	Micron Technology	MU
23	Automatic Data Processing	ADP
24	Texas Instruments	TXN
25	Applied Materials	AMAT
26	VMware (delisted)	VMW
27	eBay	EBAY
28	Analog Devices	ADI
29	Lam Research	LRCX
30	Verizon Communications	VZ

5.2 Data File Inventory

**Table 10: Attribute details for earnings_dates_cache.csv
(historical earnings dates and surprise data).**

Column	Dtype	Non-Null
ticker	str	689/689
earnings_date	str	689/689
eps_estimate	float64	687/689
reported_eps	float64	689/689
surprise_pct	float64	688/689

Table 9: Overview of all data files used and generated in the study.

Filename	Type	Shape	I/O	Source
sec_filings.csv <i>Raw SEC Form 4 insider trading filings</i>	CSV	(717, 17)	Input	OpenInsider.com
market_returns.csv <i>Daily NASDAQ Composite index returns</i>	CSV	(374, 2)	Input	Yahoo Finance ('IXIC)
portfolio_returns.csv <i>Daily returns for all 30 portfolio stocks</i>	CSV	(374, 31)	Input	Yahoo Finance
earnings_dates_cache.csv <i>Historical earnings dates and surprise data</i>	CSV	(689, 5)	Input	Yahoo Finance
managers.csv <i>Unique (ticker, insider) pairs for LLM prompting</i>	CSV	(218, 4)	Generated	01_exploratory_data_analysis.ipynb
unique_event_filings.csv <i>Deduplicated events (one per filing date, ticker, type)</i>	CSV	(492, 4)	Generated	01_exploratory_data_analysis.ipynb
events_with_overlap_flags.csv <i>Events with overlap flags for each CAR window</i>	CSV	(492, 13)	Generated	01_exploratory_data_analysis.ipynb
event_study_metadata.csv <i>Event study setup: OLS parameters per event</i>	CSV	(492, 5)	Generated	02_traditional_ar_calculation.ipynb
event_study_abnormal_returns.csv <i>Daily abnormal returns for all events (traditional)</i>	CSV	(19474, 4)	Generated	02_traditional_ar_calculation.ipynb
cumulative_abnormal_returns_exclude.csv <i>CARs for traditional & LLM (exclude mode)</i>	CSV	(840, 24)	Generated	03_car_calculation.ipynb
cumulative_abnormal_returns_both.csv <i>CARs for traditional & LLM (both mode)</i>	CSV	(984, 24)	Generated	03_car_calculation.ipynb
cumulative_abnormal_returns_merge.csv <i>CARs for traditional & LLM (merge mode)</i>	CSV	(912, 24)	Generated	03_car_calculation.ipynb
cumulative_abnormal_returns_*_clean.csv <i>Cleaned CARs (overlapping events removed)</i>	CSV	varies	Generated	03_car_calculation.ipynb
cross_sectional_results.csv <i>Descriptive stats, GRANK & comparison tests</i>	CSV	(144, 24)	Generated	04_cross_sectional_analysis.ipynb
llm_abnormal_returns.csv <i>Daily ARs from LLM with error diagnostics</i>	CSV	(19474, 16)	Generated	02_llm_ar_computation.py
llm_extracted_events_manager_*.csv <i>LLM-extracted events (manager-level prompting)</i>	CSV	(1176, 8)	Generated	00_llm_event_extraction.py
llm_extracted_events_company_*.csv <i>LLM-extracted events (company-level prompting)</i>	CSV	(212, 8)	Generated	00_llm_event_extraction.py
llm_extraction_evaluation_*.csv <i>Per-ticker extraction accuracy metrics</i>	CSV	(30, 14)	Generated	00_llm_event_extraction.py
earnings_proximity_results.csv <i>Events with earnings proximity & merged CARs</i>	CSV	(492, 15)	Generated	05_visualization.ipynb

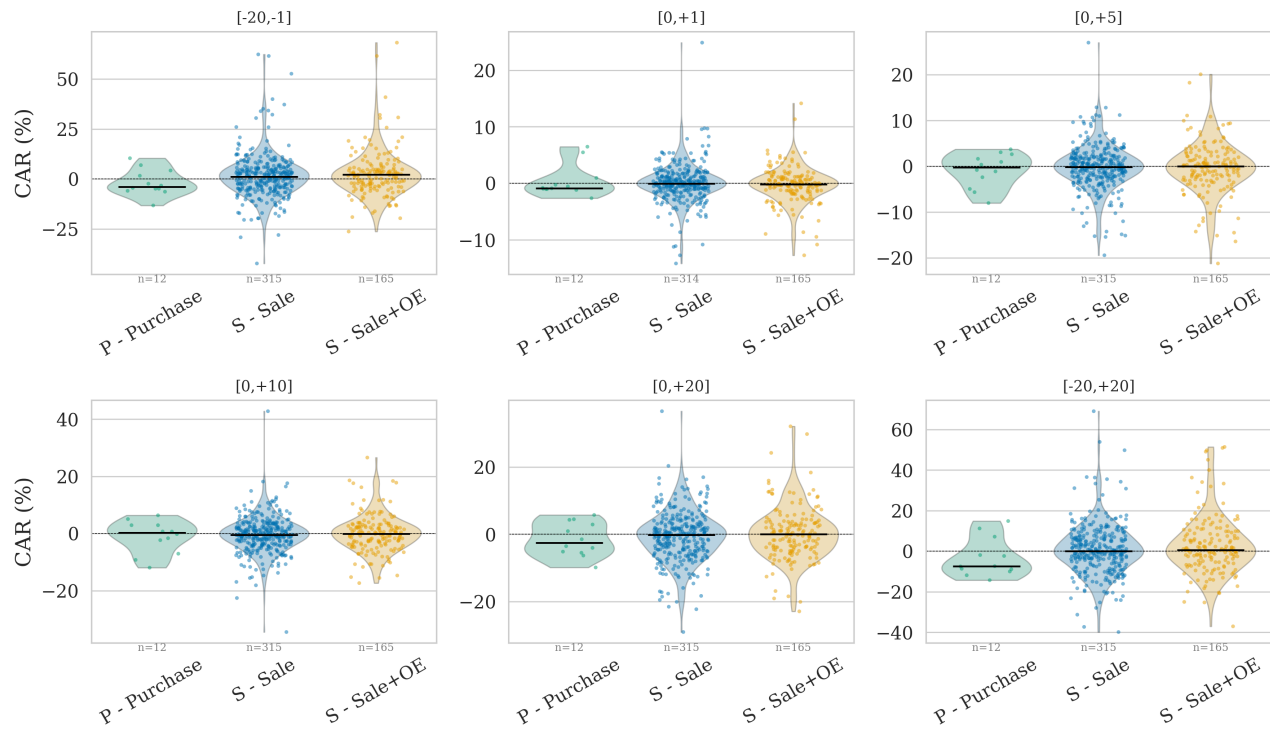


Figure 7: Density distributions of CARs by trade type and event window, showing the heavy right tails that produce significant mean CARs despite insignificant median tests.

5.3 LLM Prompts

All LLM interactions use the OpenAI-compatible chat API with temperature=0 and seed=42. Curly-brace placeholders (e.g. {ticker}) are substituted at runtime.

5.3.1 Event Extraction (00_llm_event_extraction.py). **System message** (shared by both extraction approaches):

You are a financial data assistant with knowledge of SEC Form 4 filings.
Return ONLY valid JSON. No explanation or markdown.

Company-level user prompt (30 prompts, one per ticker):

List ALL director dealings (insider trading) transactions for {company_name} ({ticker}) between November 1, 2024 and May 31, 2025. Include purchases, sales, and option-exercise sales.

For each transaction, return: date (YYYY-MM-DD), insider_name, role, transaction_type (one of: "P - Purchase", "S - Sale", "S - Sale+OE"), shares, price_per_share, total_value.

Return ONLY a JSON array of objects. No explanation or markdown.

Manager-level user prompt (218 prompts, one per (insider, ticker) pair):

List ALL director dealings (insider trading) transactions performed by {insider_name} at {company_name} ({ticker}) between November 1, 2024 and May 31, 2025.

For each transaction, return: date (YYYY-MM-DD), insider_name, role, transaction_type (one of: "P - Purchase", "S - Sale", "S - Sale+OE"), shares, price_per_share, total_value.

Return ONLY a JSON array of objects. No explanation or markdown.

5.3.2 Abnormal Return Computation (02_llm_ar_computation.py).

System message (shared by both prompt variants):

You are a financial analyst. Perform the calculations and return ONLY a JSON object with the numeric results. No explanation.

Abnormal Return Calculation Prompt (up to ~19,000 calls; used in all reported results):

Perform an event study Abnormal Return calculation.

****Step 1 – Estimate Market Model parameters using OLS regression on this estimation window data:****

Day | Stock_Return | Market_Return
{estimation_window_table}

Regression model: $\text{Stock_Return} = \alpha + \beta \times \text{Market_Return} + \epsilon$

****Step 2 – Calculate the Abnormal Return for this single trading day:****

Stock_Return: {stock_return}
Market_Return: {market_return}

Expected_Return = $\alpha + \beta \times \text{Market_Return}$
AR = Stock_Return - Expected_Return

****Return JSON only:****
{ "alpha": <number>, "beta": <number>, "expected_return": <number>, "ar": <number> }